

O2N Engine API Guide

Postman testing, endpoint payloads and responses, plus a frontend implementation prompt

Local Server

Start the backend before testing. The API base URL in this guide is `http://127.0.0.1:8081`.

```
cd E:\hackthon\OpenAi_Hackathone_Backend\backend
python -m uvicorn app.main:app --host 127.0.0.1 --port 8081 --reload
```

Swagger documentation: <http://127.0.0.1:8081/docs>

Important Path Rule

The project path is read by the backend machine. For this local setup, use a real absolute Windows path such as `E:\projects\legacy-shop`. A hosted browser application cannot access a visitor's disk path without a ZIP upload or desktop integration.

Postman Collection

Import `O2N_Engine_API.postman_collection.json` in Postman: Import -> File -> select the collection. Then update collection variables: `baseUrl`, `projectPath`, and `outputDirectory`. `outputDirectory` must be a new path outside `projectPath`.

Endpoint Summary

Method	Endpoint	Purpose
GET	/	Backend startup status
GET	/health	Health check
GET	/api/v1/conversion-targets	Target stack dropdown values
POST	/api/v1/projects/analyze	Analyze a codebase and get roadmap
POST	/api/v1/projects/analyze/github	Clone and analyze a public GitHub repository
POST	/api/v1/projects/convert	Generate converted files into a new folder

1. GET /

Use this optional endpoint to verify that the backend is running.

GET `http://127.0.0.1:8081/`

```
200 OK
{
  "message": "O2N Engine Backend Running",
  "status": "success",
  "docs": "/docs",
  "health": "/health"
}
```

2. GET /health

Use before enabling Analyze or Convert actions in the frontend.

```
GET http://127.0.0.1:8081/health
```

```
200 OK
{
  "status": "ok"
}
```

3. GET /api/v1/conversion-targets

Use this response for a target technology dropdown. The frontend should additionally allow a custom target text value.

```
GET http://127.0.0.1:8081/api/v1/conversion-targets
```

```
200 OK
{
  "targets": ["Next.js + FastAPI", "React + Node.js/Express", "React + Django", "Vue + FastAPI", "Angular + Spring Boot",
  "custom_target_supported": true
}
```

4. POST /api/v1/projects/analyze/github

Use when the user chooses a public GitHub repository instead of a local folder. The backend accepts only HTTPS github.com owner/repository URLs, clones the repository to its managed workspace, then runs the normal analysis flow. Private repositories need GitHub OAuth/App support and are not supported yet.

```
{
  "name": "Public GitHub Repository",
  "repository_url": "https://github.com/owner/repository",
  "target_stack": "Next.js + FastAPI"
}
```

200 OK response includes:

```
{
  "source_type": "github_repository",
  "repository_url": "https://github.com/owner/repository",
  "path": "...\\repository_workspace\\repository_a1b2c3d4"
}
```

400 invalid URL:

```
{
  "detail": "Use a public HTTPS GitHub URL such as https://github.com/owner/repository."
}
```

5. POST /api/v1/projects/analyze

Analyzes supported source files, counts languages, runs optional syntax and Semgrep scans, detects PHP/jQuery/Python 2 patterns, and returns a roadmap. The call is synchronous, so show a loading state.

Payload

```
{
  "name": "Legacy Shop",
  "path": "E:\\projects\\legacy-shop",
  "target_stack": "Next.js + FastAPI"
}
```

Required fields: name (1-120 chars), path (existing absolute backend-readable directory). target_stack is optional and defaults to Next.js + FastAPI.

Successful response: 200 OK

```
{
  "id": "b92cae61-78bf-48c8-96b7-5f8b1e335baf",
  "name": "Legacy Shop",
  "path": "E:\\projects\\legacy-shop",
  "status": "complete",
  "created_at": "2026-07-20T12:30:00Z",
  "languages": {"PHP": 18, "JavaScript": 12},
  "findings": [{"severity": "medium", "category": "syntax", "message": "Tree-sitter found a syntax error", "file": "E:\\p"},
  "recommendations": [{"current_technology": "PHP", "reason": "PHP source files were detected; modern API boundaries can s"},
  "roadmap": [{"phase": "1. Baseline", "action": "Add tests and capture current behavior", "rationale": "Protect business"},
  "summary": "Scanned 30 source files across 2 language(s)."}
}
```

Error response: 400 Bad Request

```
{
  "detail": "The provided project path does not exist or is not a folder."
}
```

6. POST /api/v1/projects/convert

Converts selected project-relative files or folders with Claude AI. ANTHROPIC_API_KEY is required. The source project is never overwritten; generated files are written only to a new output directory.

Payload

```
{
  "name": "Legacy Shop Migration",
  "path": "E:\\projects\\legacy-shop",
  "source_paths": ["api/users.php", "public/js/app.js"],
  "target_stack": "Next.js + FastAPI",
  "output_directory": "E:\\converted-projects\\legacy-shop-next-fastapi"
}
```

source_paths must be relative to path. A directory is allowed, but no more than 20 supported source files may be expanded. output_directory is optional, must be outside path, and must not exist already.

Successful response: 200 OK

```
{
  "id": "9c1e977f-6f4f-4d6d-a943-9af1126c4eb4",
  "name": "Legacy Shop Migration",
  "project_path": "E:\\projects\\legacy-shop",
  "target_stack": "Next.js + FastAPI",
  "output_directory": "E:\\converted-projects\\legacy-shop-next-fastapi",
  "generated_files": [{"path": "backend/app/routes/users.py", "content": "from fastapi import APIRouter..."}],
  "findings": [],
  "summary": "Generated 1 file(s) from 2 selected source file(s)."
}
```

Common conversion errors

```
{ "detail": "ANTHROPIC_API_KEY is required to generate converted code." }
{ "detail": "Source path does not exist: api/missing.php" }
{ "detail": "Source path escapes the project directory: ../private-file.py" }
{ "detail": "Output directory already exists; choose a new empty location." }
```

Frontend Integration Prompt

Copy the following prompt into your frontend coding assistant.

Build a responsive React or Next.js frontend for the O2N Engine API at <http://127.0.0.1:8081>. Do not show raw JSON.

Create these flows:

1. On load, call GET /health and GET /api/v1/conversion-targets.
2. Add a source selector: Local Folder or Public GitHub Repository.
3. Local form: name, absolute project path, target stack dropdown, and custom target input. Submit POST /api/v1/projects/analyze/local.
4. GitHub form: name, public HTTPS GitHub URL, and target stack. Submit POST /api/v1/projects/analyze/github.
5. Report UI: language counts, findings grouped by severity, selectable legacy recommendations, roadmap steps, summary, and findings.
6. Conversion form: returned project path, selected project-relative source paths, target stack, optional separate output directory.
7. Conversion result: output directory, generated-file sidebar, selected file code preview, findings, error state, and load status.

Use the response shapes from this guide. Show backend detail messages on non-2xx responses. Disable submit controls during analysis.

Configuration

```
ANTHROPIC_API_KEY=your_key_here
MONGODB_URI=mongodb://localhost:27017
MONGODB_DATABASE=o2n_engine
CORS_ORIGINS=http://localhost:3000,http://localhost:5173
```

Without ANTHROPIC_API_KEY, analysis returns a fallback roadmap but code conversion returns 400. MongoDB and static-analysis tools are optional; unavailable services do not stop the core analysis response.